# 🧩 Topic 5: Unification in FOL

### 1. What is Unification?

- **Definition:** Unification is the process of making two different logical expressions look **exactly identical**.
- **How is it done?** It is done using a process called **"Substitution"**. We replace unknown variables (like $x, y$) with specific constants or functions so that both sentences match perfectly.
- **Substitution Set:** The changes we make are written in brackets. E.g., if we replace $x$ with $a$, we write it as `[a/x]`.

### 2. The 3 Basic Conditions for Unification

Before you can even try to unify two expressions, they MUST pass these 3 strict rules:

1. **Same Predicate Symbol:** Both expressions must have the exact same starting word (predicate).
   - *Example:* `Prime(x)` and `Prime(11)` can unify. But `Prime(x)` and `Even(11)` CANNOT unify.
2. **Same Number of Arguments:** Both expressions must have the exact same number of items inside the brackets `()`.
   - *Example:* `P(x, y)` has 2 arguments. `P(a, b, c)` has 3 arguments. They CANNOT unify.
3. **No Self-Reference:** If two similar variables are present in the same expression and conflict, unification fails.

### 3. The Unification Algorithm

This is the step-by-step logic the machine follows. Let's call our two expressions **A1** and **A2**:

- **Step 1:** Check the basics.

    - If the predicates are NOT the same $\rightarrow$ **FAIL**.
    - If the number of arguments are different $\rightarrow$ **FAIL**.
- **Step 2:** If A1 or A2 is a variable/constant:
    - If A1 and A2 are already identical $\rightarrow$ return NIL (do nothing, they already match).
    - Else, substitute A2 in A1 (or A1 in A2) to make them match.
- **Step 3:** Keep looping through every argument inside the brackets one by one until everything matches.
- **Step 4:** Return the final **SUBSTITUTION SET**.

### 4. Examples (How to actually solve it)

#### Example 1: The Simple One

- **Given:** `Prime(11)` and `Prime(y)`
- **Step 1:** Predicates are same ( `Prime` ). Arguments are same (1 each).
- **Step 2:** We need them to match perfectly. So, we replace the variable `y` with the constant `11` .
- **Substitution Set:** `[11/y]` .
- **Result:** Both become `Prime(11)` . **Unified Successfully!**

#### Example 2: The Complex One

- **Given A1:** `Q(a, g(x, a), f(y))`
- **Given A2:** `Q(a, g(f(b), a), x)`
- **Solving step-by-step:**
    1. *First terms:* `a` and `a` . (Already match).
    2. *Second terms:* `g(x, a)` and `g(f(b), a)` . To make them match, we must substitute $x$ with $f(b)$.

    - **Substitution 1:** `[f(b)/x]` .
    - *Update the whole formula:* Replace every $x$ with $f(b)$.
    - New A1: `Q(a, g(f(b), a), f(y))`
    - New A2: `Q(a, g(f(b), a), f(b))` *(Notice the last term $x$ became $f(b)$).*
  3. *Third terms:* `f(y)` and `f(b)` . To make them match, we must substitute $y$ with $b$.
    - **Substitution 2:** `[b/y]` .
    - *Update the whole formula:*
    - Final A1: `Q(a, g(f(b), a), f(b))`
    - Final A2: `Q(a, g(f(b), a), f(b))`
- **Result:** Both formulas are now 100% identical. **Unified Successfully!**